

Compound Parabolic Concentrator Cross-Sections and 3D-Printed Designs for Solar Thermophotovoltaic Systems

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Background

Shockley-Queisser limit. Single-junction silicon solar cells are limited to ~31% efficiency due to spectral mismatch. [1].

Solar Thermophotovoltaic systems STPVs use an absorber and emitter to convert sunlight into thermal radiation better matched to a PV cell, with theoretical efficiencies up to 85%. With this efficiency, an area the size of New Jersey, could power Earth's entire population and industries. Recent STPV systems have achieved as much as 42% efficiency through the utilization of Class III-V PV cells, and selective emitters, however these systems can be impractical and expensive due to low availability of materials.

Methods

Understanding the system. A solar thermophotovoltaic (STPV) system converts sunlight into heat and re-emits it as infrared radiation matched to the PV cell bandgap. Concentrated sunlight is absorbed as heat, then emitted toward the PV cell.

Concentrator. Focuses sunlight on to the absorber. This work studies Compound Parabolic Concentrators (CPCs).

Absorber. Converts concentrated sunlight into heat.

Emitter. Radiates thermal energy towards PV cell at wavelengths close to bandgap.

PV Cell. Semiconductors that absorb photons above their bandgap to convert to energy into electricity.

Heat Sink. Dissipates heat from PV cell to maintain efficiency.

Concentrator Sweep. We conducted a parameter sweep over the shape of the concentrator cross-section: circle, ellipse, rectangle; acceptance angle; and relevant dimensions. Introducing split, a new dimension that allows the concentrator to be lengthened.

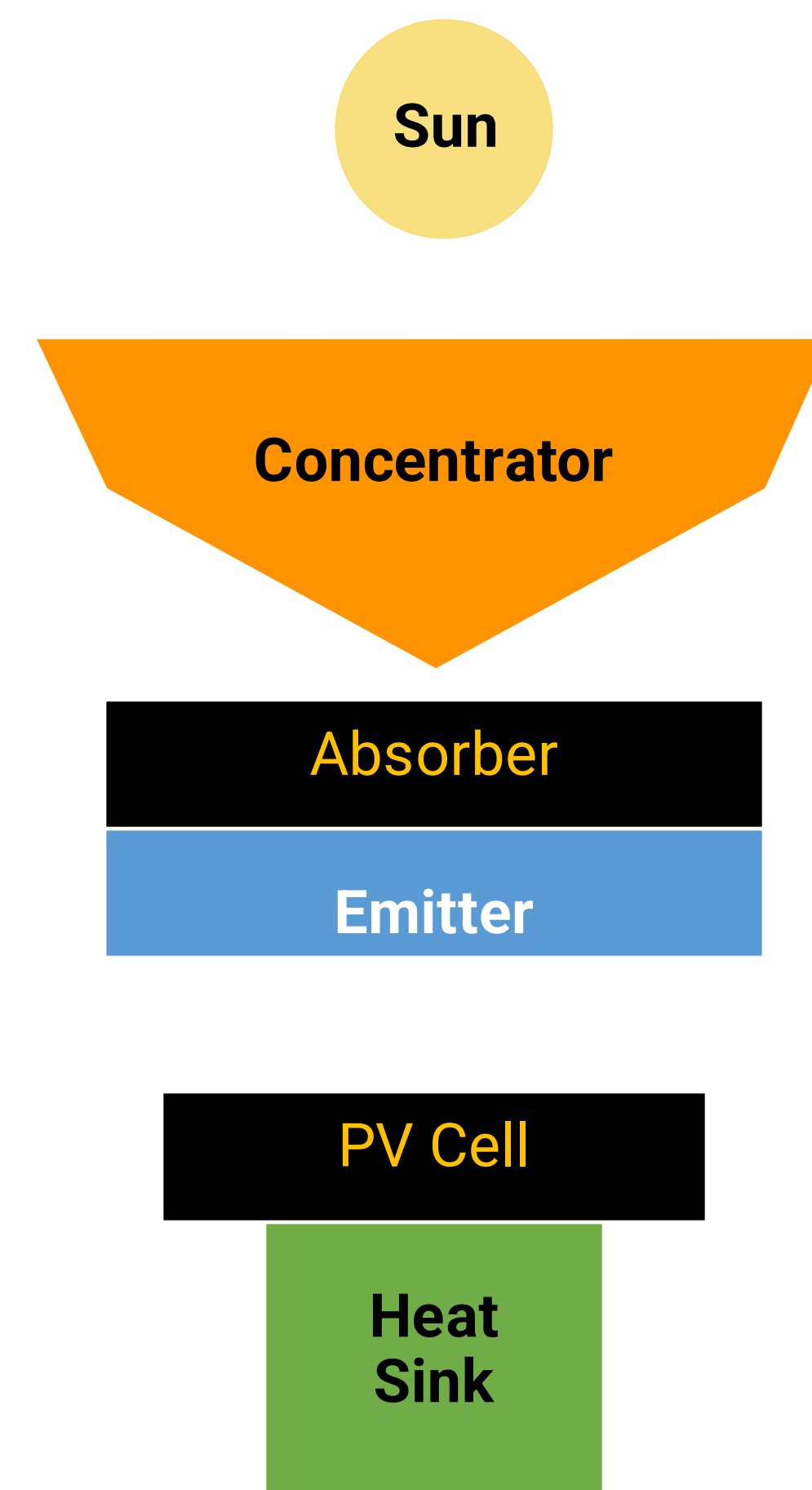


Figure 1. Components of an STPV System.

Build

Absorber / Emitter Combination:
Figure 2. Absorber / Emitters Tested

Silicon Carbide (SiC / Ceramic)
3 x 3 in

Black Silicon Carbide (SiC / Ceramic)
3 x 3 in

Silicon Carbide (SiC / Ceramic) 1 x 1 cm

Absorber painted black; Silicon Carbide (SiC / Ceramic) 1 x 1 cm

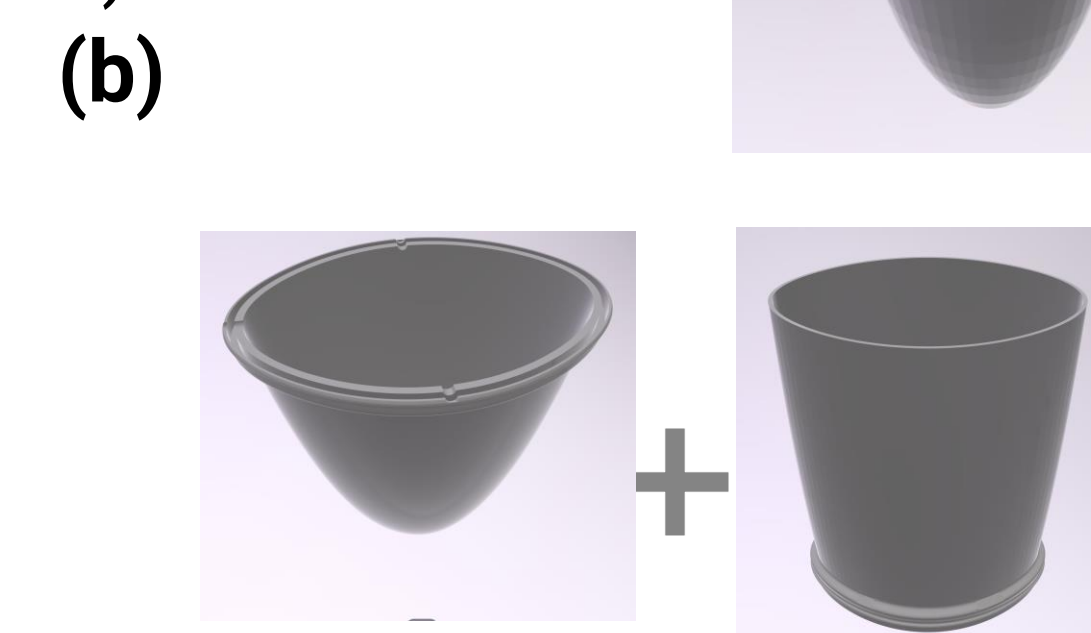
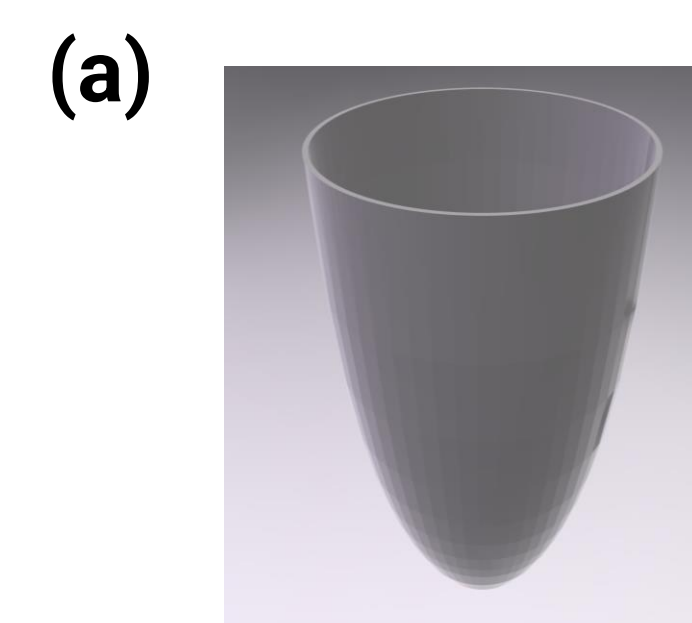
One side black finish Aluminum 1 x 1 cm

Assembly Rig: CAD with Autodesk Fusion

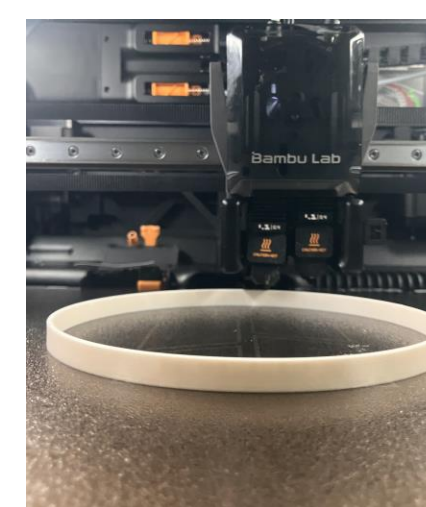
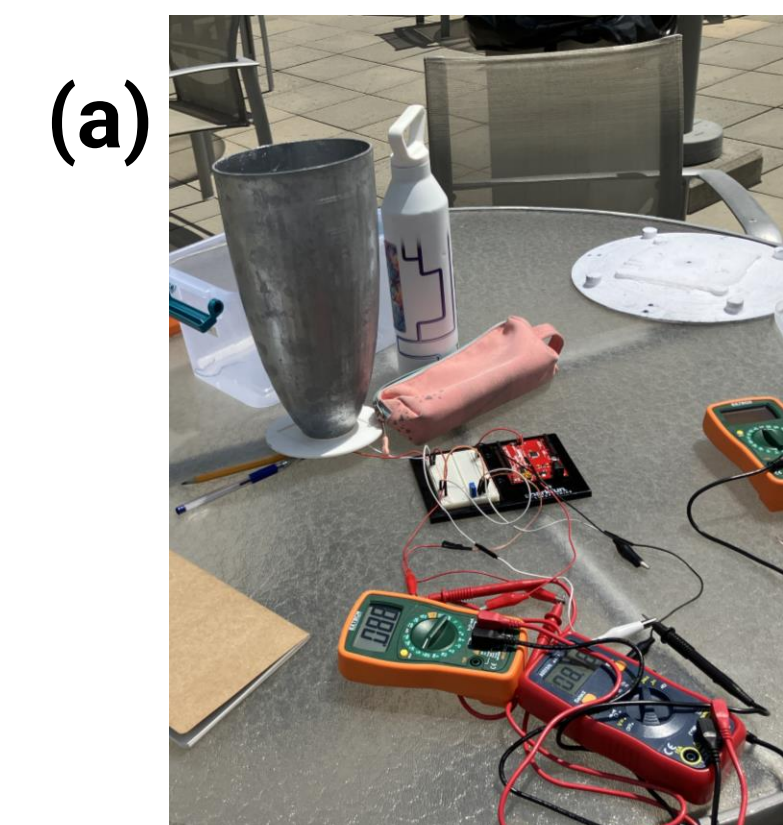


Figure 3. Variable heights of Assembly Rigs, shown in CAD

Concentrators:
CAD with Cadquery library on python (with Claude)



CAD



Print

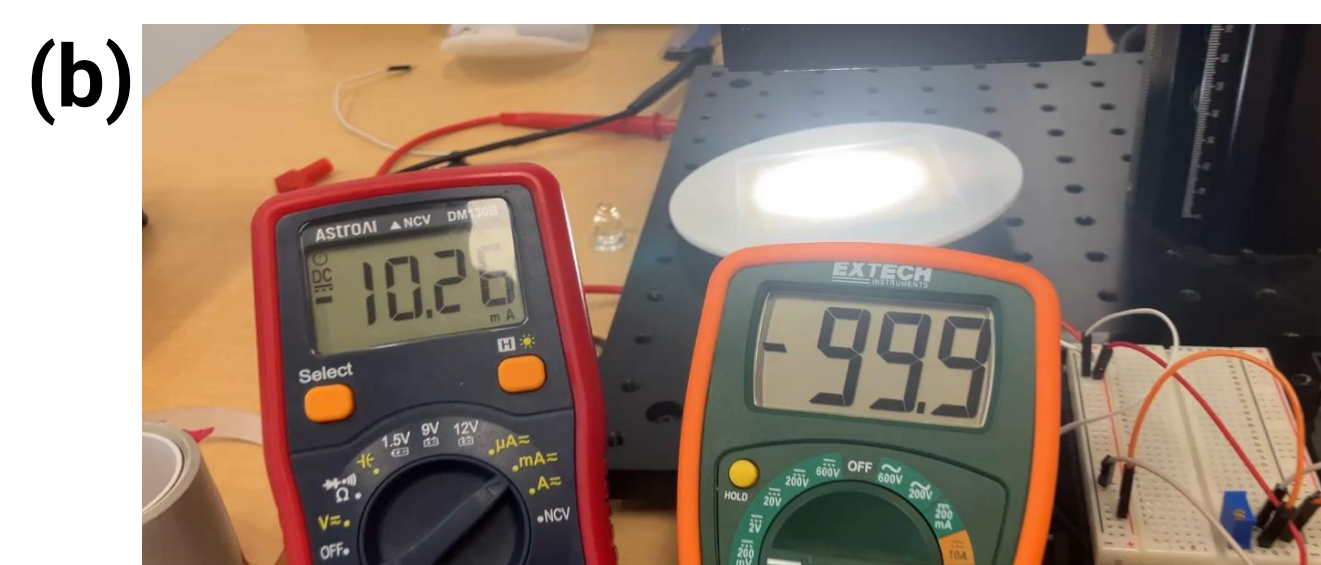


Figure 5. (a) Outdoor Testing Setup. (b) Indoor Testing setup

Paint



Results

Outdoor Testing

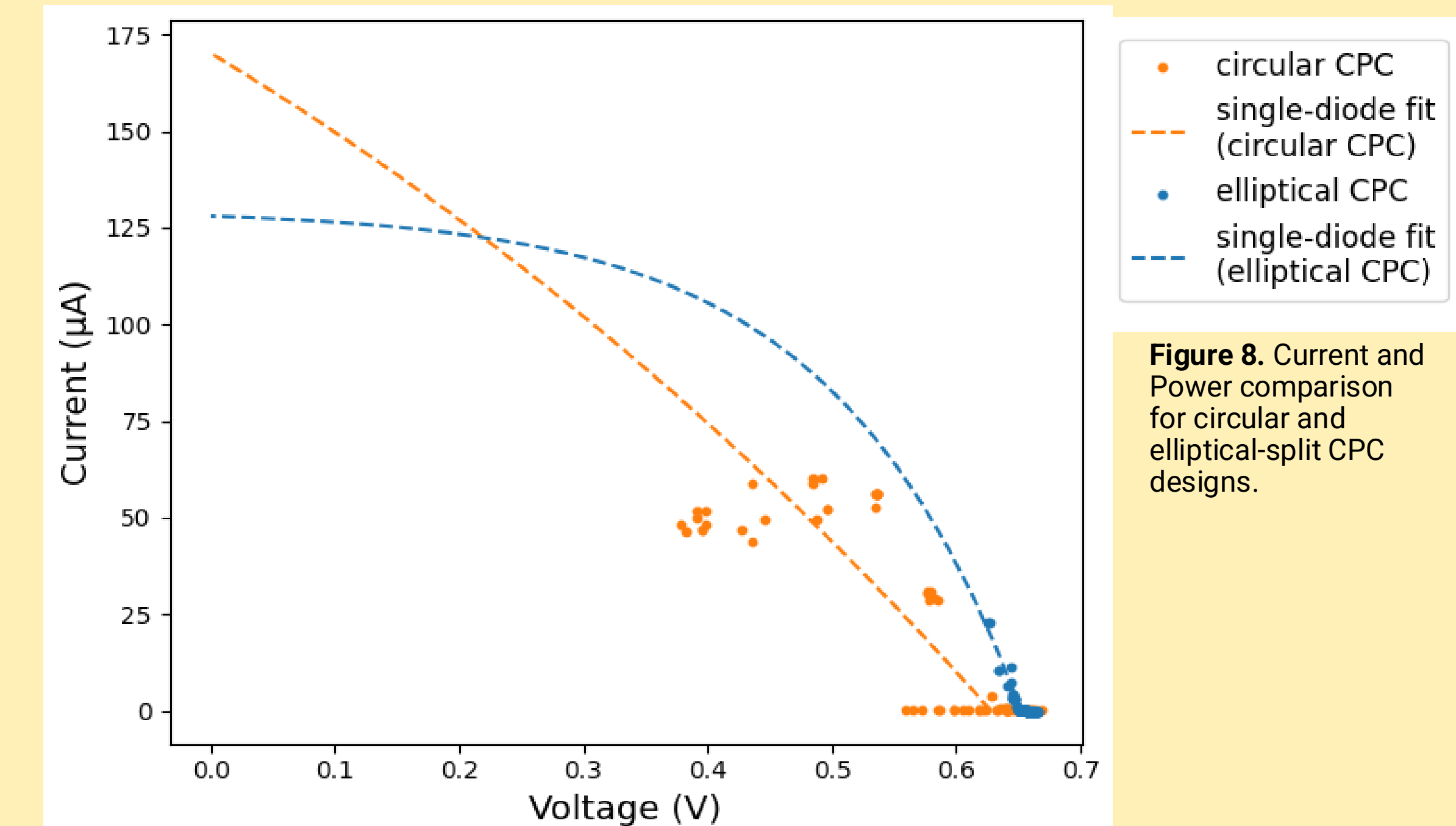
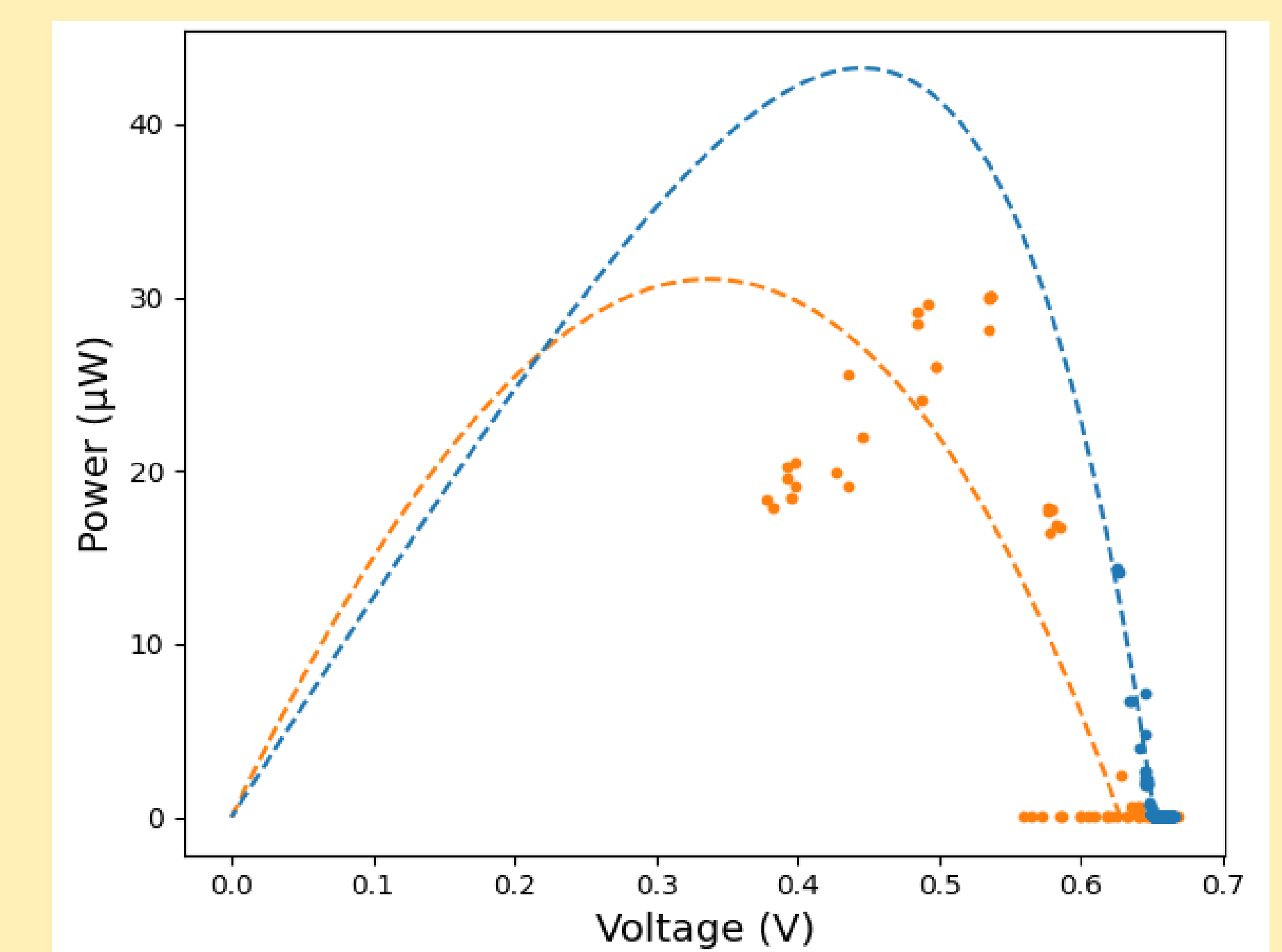


Figure 8. Current and Power comparison for circular and elliptical-split CPC designs.



Under optimized resistance the elliptical CPC could achieve an efficiency as high as 2.1% compared to 0.36% for the circular CPC, and a maximum power output of 43.25 mW compared to 31.1 mW

Indoors Testing

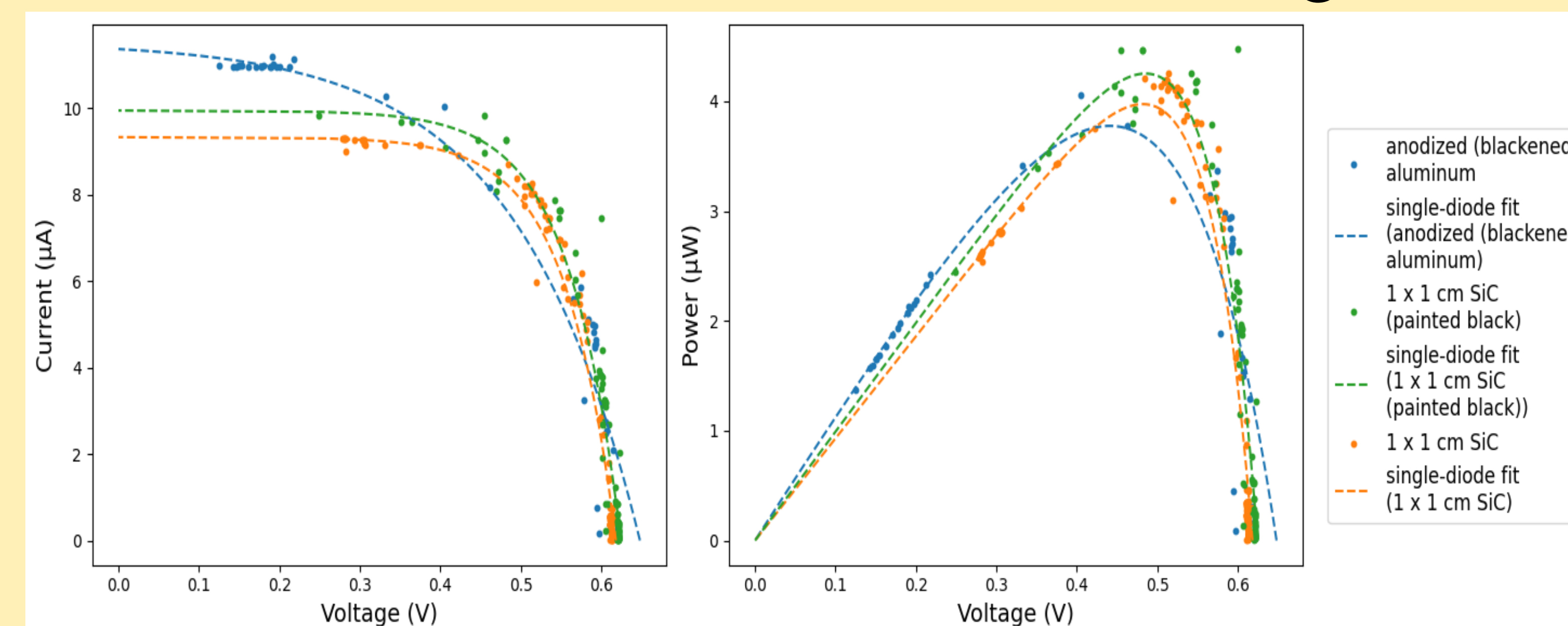
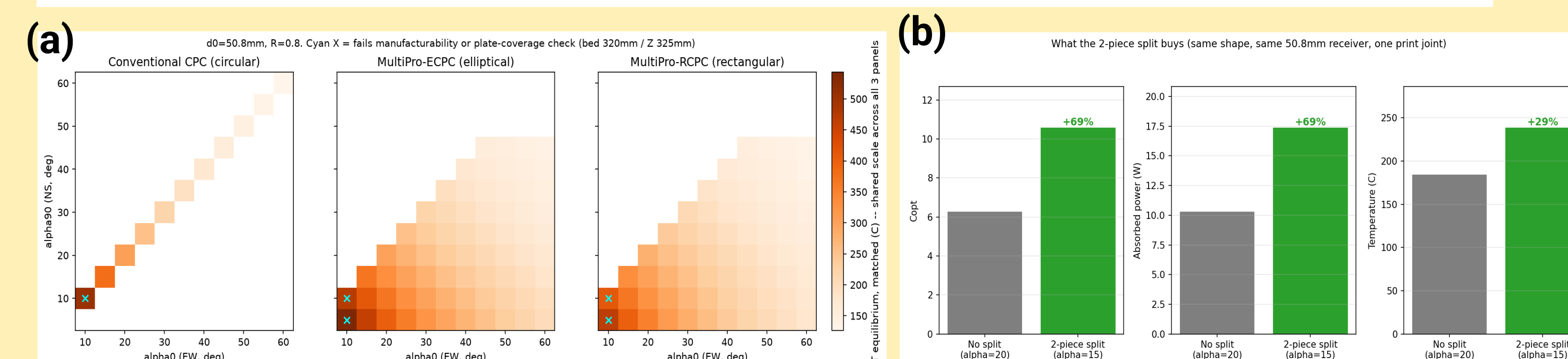


Figure 6. All 1 x 1 cm cut materials exhibited comparable performance from 3.5 to 4% efficiency, with a maximum power output of about 4 mW. The larger sheets (76.2 x 76.2 mm) outputted no voltage and no currents due to difficulty heating them.

Figure 7. Sweeps run on different cross-sectional shape designs for Solar concentrators. (a) Analyzing Temperature as a function of acceptance angle. (b) Effect of Split design on efficiency



Acceptance Angle. A large acceptance angle increases ability to use the design throughout the day, making it more practical for usage, but it also reduces efficiency. Elliptical designs, able to maintain a wide East-West angle alongside a narrow North-South angle, showed more resistance to this change.

Split. For both circular and elliptical CPCs, a 2-piece split print was predicted to increase system efficiency by 69%.

Discussion

Novelty. Previous 3D-printed CPCs were limited to circular cross-sections. The usage of a male-female interlocking geometry enables printing a longer elliptical CPC shown to achieve a theoretical efficiency of 2.1% under optimized resistance for a silicon PV cell-anodized aluminum absorber/emitter system.

Limitations. The system used a silicon PV cell instead of the higher-efficiency InGa cells typically used in STPV systems due to cost constraint. In addition, the emitter-PV cell spacing was not optimized, reducing radiative coupling efficiency. Finally, the chrome spray coating (~50% reflectivity) lowered concentrator optical efficiency.

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References

Scan this QR code for references.

